NCERT Summary: Topic 04: Major Domains & Major Landforms of the Earth — Raw Notes

Source Integration

- NCERT Class 6 Geography: The Earth: Our Habitat, Chapter 5: Major Domains of the Earth
- NCERT Class 6 Geography: The Earth: Our Habitat, Chapter 6: Major Landforms of the Earth

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Core Factual & Conceptual Architecture

1. The Four Major Earth Domains

- Etymological Roots:
  - Lithosphere: From Greek Lithos (Stone), representing the solid crustal portion of the Earth.
  - Atmosphere: From Greek Atmos (Vapour), representing the enveloping gaseous mantle.
  - Hydrosphere: From Greek Hudor (Water), representing all water phases on the planet.
  - Biosphere: From Greek Bios (Life), representing the narrow intersection zone supporting life.
- Domain Interdependence: The biosphere exists exclusively at the dynamic contact boundary of the lithosphere, hydrosphere, and atmosphere. Disruption in one sphere triggers cascading degradation across the others (e.g., deforestation accelerates soil erosion and atmospheric CO_2 buildup).

2. Lithosphere & Global Continental Distribution

- Structural Composition: Solid rocks of the Earth's crust and the overlying thin mantle of nutrient-bearing topsoil sustaining vegetation.
- Major Surface Divisions: Continental landmasses and expansive oceanic basins.
- Elevation Benchmarks: Measured against mean sea level (datum zero). Highest terrestrial summit: Mount Everest (8,848 m). Deepest oceanic depression: Mariana Trench in the Pacific Ocean (11,022 m).
- Historic Mountaineering Benchmarks:
  - Edmund Hillary (New Zealand) and Tenzing Norgay Sherpa (India): First successful ascent of Mount Everest on 29 May 1953.
  - Junko Tabei (Japan): First woman to summit Everest on 16 May 1975.
  - Bachendri Pal: First Indian woman to summit Everest on 23 May 1984.
- The Seven Continents:
  - Asia: Largest continent, occupying approximately one-third of total terrestrial land area. Located in the Eastern Hemisphere; crossed by the Tropic of Cancer. Separated from Europe by the Ural Mountains. Combined landmass is called Eurasia.
  - Europe: Positioned west of Asia, bounded by water on three sides; traversed by the Arctic Circle.
  - Africa: Second largest continent. Unique in having the Tropic of Cancer, Equator, and Tropic of Capricorn traverse its landmass. Contains the Sahara Desert (largest hot desert) and the Nile River (longest river globally).
  - North America: Third largest continent, positioned entirely in the Northern and Western Hemispheres; bordered by three oceans. Joined to South America via the narrow Isthmus of Panama.
  - South America: Positioned largely in the Southern Hemisphere; hosts the Andes (longest mountain chain globally) and the Amazon River (largest river by water

volume).
    - Australia: Smallest continent, lying entirely in the Southern Hemisphere; termed an island continent due to complete oceanic encirclement.
    - Antarctica: Sits entirely in the Southern Hemisphere with the South Pole at its center; permanently blanketed by thick continental ice sheets. Houses scientific research outposts, including Indian stations Maitri and Dakshin Gangotri.

3. Hydrosphere & Oceanic Basin Systems

- Surface Distribution: 71 percent of the planet is submerged by water, while 29 percent comprises dry landmasses.
- Saline vs. Fresh Water: Over 97 percent of Earth's water resides in saline oceans, unusable for direct human consumption. Most of the remaining 3 percent is trapped in ice sheets, mountain glaciers, or deep aquifers, leaving a tiny fraction as accessible surface freshwater.
- Primary Oceanic Dynamics: Driven by three continuous movements: waves, tides, and planetary ocean currents.
- The Five Oceanic Basins:
    - Pacific Ocean: Largest oceanic basin covering one-third of the globe; roughly circular; contains the Mariana Trench; flanked by Asia, Australasia, and the Americas.
    - Atlantic Ocean: Second largest; distinct S-shaped configuration; flanked by the Americas on the west and Europe/Africa on the east. Features a highly indented coastline that creates natural harbors, making it the most commercially active ocean.
    - Indian Ocean: Sole ocean named after a nation; roughly triangular; bounded north by Asia, west by Africa, and east by Australia.
    - Southern Ocean: Encircles the Antarctic landmass, extending northward to the 60 degrees South latitude line.
    - Arctic Ocean: Smallest ocean; positioned within the Arctic Circle around the North Pole; linked to the Pacific Ocean via the shallow Bering Strait; flanked by northern Eurasia and North America.

4. Atmosphere: Layering, Chemistry & Pressure

- Vertical Extent: Gaseous envelope extending upward to approximately 1,600 kilometers.
- Five Atmospheric Strata (Surface Upward): Troposphere, Stratosphere, Mesosphere, Thermosphere, and Exosphere.
- Volumetric Chemical Composition (Clean, Dry Air):
    - Nitrogen: 78 percent; sustains plant tissue synthesis and cellular development.
    - Oxygen: 21 percent; indispensable for biological respiration ("breath of life").
    - Trace Gases (CO2, Argon, Water Vapour, etc.): 1 percent combined.
    - Carbon Dioxide Role: Absorbs terrestrial longwave heat radiation, keeping the planetary surface warm through the natural greenhouse effect, while supporting floral photosynthesis.
- Pressure and Density Gradients: Both air density and atmospheric pressure decline rapidly with increasing altitude, necessitating supplemental oxygen for high-altitude mountaineers. Horizontal barometric pressure differences drive air movement from high-pressure cells toward low-pressure cells, generating wind.

5. Geomorphic Processes Shaping Earth's Relief

- Dual Geomorphic Forces:
    - Endogenic (Internal) Processes: Sub-crustal convective forces driving continuous tectonic uplift, faulting, and crustal subsidence.
    - Exogenic (External) Processes: Continuous sculpturing via surface denudation: degradation through weathering/erosion and aggradation through sedimentary deposition, driven by running water, moving ice, and wind.

6. Typology of Major Landforms

- Mountain Systems:
  - Definition: Natural elevations of the crust rising sharply above surroundings, featuring narrow summits and broad bases. Carry permanent ice reservoirs termed glaciers. Include submerged marine volcanic summits, such as Mauna Kea in Hawaii (10,205 m tall from ocean floor, exceeding Everest).
  - Fold Mountains: Formed via crustal compression. Young fold ranges feature steep rugged peaks and deep valleys (Himalayas, Alps). Old fold mountains are rounded and denuded by erosion (Aravalli Range in India, Appalachian Mountains in North America, Ural Mountains in Russia).
  - Block Mountains: Formed when expansive crustal blocks fracture and displace vertically along faults. Uplifted blocks are termed Horsts; down-dropped sunken rift blocks are termed Graben (e.g., the Rhine Valley and Vosges Mountains in Europe).
  - Volcanic Mountains: Formed by the accumulation and solidification of ejected volcanic pyroclasts and lava (e.g., Mount Kilimanjaro in Africa, Mount Fujiyama in Japan).
  - Ecological & Human Utilities: Headwaters for perennial river systems, hydroelectric potential, terrace farming, timber, resins, and alpine adventure tourism.
- Plateaus (Tablelands):
  - Definition: Uplifted, flat-topped tablelands standing abruptly above adjacent terrain, bound by steep escarpments.
  - Global Examples: Deccan Plateau of India (ancient basaltic shield); East African Plateau (Kenya, Uganda, Tanzania); Western Plateau of Australia; Tibetan Plateau (world's highest tableland, 4,000 to 6,000 m elevation).
  - Economic Value: Rich mineral reserves (gold and diamonds in African plateaus; coal, iron ore, and manganese across the Chhotanagpur Plateau in India). Escarpments host major waterfalls (Hundru Falls on the Subarnarekha River, Jog Falls in Karnataka). Basaltic lava plateaus provide fertile black regur soils ideal for cash cropping.
- Plains (Depositional Lowlands):
  - Definition: Expansive tracts of flat land, generally not exceeding 200 meters above mean sea level.
  - Genesis: Constructed by rivers eroding upstream mountain ranges, carrying silt, sand, and clay, and depositing sediments across low-gradient valleys.
  - Global Examples: Indo-Gangetic Plains of India (Ganga and Brahmaputra rivers) and the Yangtze River Plains of China.
  - Human Concentration: Deep fertile alluvium, flat topography for transport grids, and abundant water make plains the most densely populated habitats globally.

✍ Notes & Updates